# Relations and Functions

special kind of relation



### Relation

A **relation** is a set of ordered pairs. Its inputs form the **domain**; its outputs form the **range**.

### Function

A **function** assigns *exactly one* output to every input. Different inputs may share the same output.

Function test: ask whether any single input is paired with more than one output. On a graph, this becomes the **vertical line test**.

- relation has domain and range/co-domain
- function work like: $f(x)=x^2$, $y=\frac{4}{x}$, $g(x)=3x-4$ ....
- domain is the numbers we can plug in, codomain are the outputs

# Find the Domain and Range



Let

$$f(x) = \frac{\sqrt{5 - 2x},}{x + 1}.$$

**a.** State the domain in interval notation.

**b.** Explain which restriction comes from the square root and which comes from the denominator.

**c.** Without graphing technology, decide whether 0 belongs to the range. Justify your answer.

a,b) what are the number can we plug into the function.

notice, $\sqrt{5-2x}$, so $5-2x \geq 0$

$-2x \geq -5$

$x \leq \frac{5}{2}$

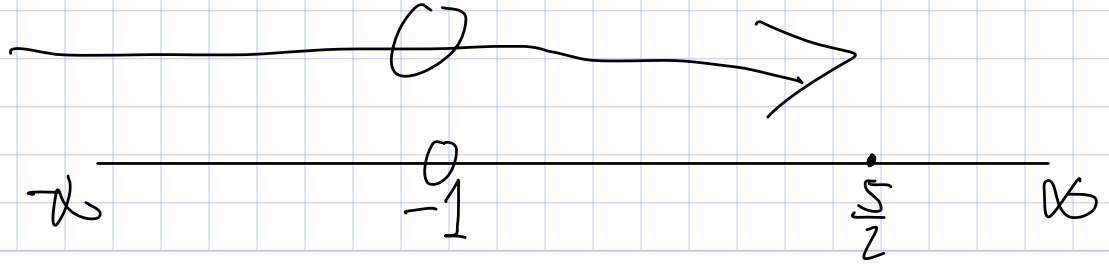
$x \leq \frac{5}{2}$

, Notice, $\frac{\cdots}{x+1}$

no zero in deno ....

$x+1 \neq 0$

$x \neq -1$




$(-\infty, -1) \cup (-1, \frac{5}{2}]$

c) find $f(a)=0$, when numerator = zero, the fraction/$f(x)$ = zero,

$$\sqrt{5-2x}=0$$

$$5-2x=0$$

$$-2x=-5$$

$$x=\frac{5}{2}$$

# Combine Two Functions—Then Track the Domain



Let $f(x) = 2x - 3$ and $g(x) = \sqrt{x+4}$.

**a.** Find $(f \cdot g)(x)$ and state its domain.

**b.** Find $\left(\frac{f}{g}\right)(x)$ and state its domain.

**c.** Why are the two domains different even though the same functions are used?

2, 3 $\quad \frac{2}{3} = ?$

$\frac{f}{g} \neq f \cdot g$ $\quad 2 \cdot 3 = 6$

(a). $(f \cdot g)(x) = f(x) \cdot g(x) = (2x-3) \cdot \sqrt{x+4}$

$f(x)$ $\quad g(x)$

$x+4 \geq 0$

$x \geq -4$

$[-4, \infty)$

(b). $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{2x-3}{\sqrt{x+4}}$

- $x+4 \geq 0$, $x \geq -4$
- $\sqrt{x+4} \neq 0$, $x \neq 4$

$\Rightarrow x > 4$, $(4, \infty)$

# Simplify a Difference Quotient



For $f(x) = x^2 - 3x + 2$, simplify

find: $\dfrac{f(x+h) - f(x)}{h}, \quad h \neq 0.$

formula: $(a+b)^2 = a^2 + 2ab + b^2$

Your final expression should not have $h$ in the denominator. Clearly show where the common factor of $h$ appears.

- $f(x)$, $h$ easy to see.
- find $f(x+h)$, replace $x$ by $x+h$.

$f(x+h) = (x+h)^2 - 3(x+h) + 2$

$= x^2 + 2xh + h^2 - 3x - 3h + 2$

$f(x)$ = $x^2 - 3x + 2$

$$\frac{f(x+h) - f(x)}{h} = \frac{x^2 + 2xh + h^2 - 3x - 3h + 2 - (x^2 - 3x + 2)}{h}$$

$$= \frac{x^2 + 2xh + h^2 - 3x - 3h + 2 - x^2 + 3x - 2}{h}$$

$$= \frac{2xh + h^2 - 3h}{h}$$

there is $h$ on each part, so factor $h$.

$$= \frac{h(2x + h - 3)}{h} = 2x + h - 3$$

# What a Graph Can Tell Us



### Zeros and intercepts

A real number $r$ is a **zero** of $f$ when $f(r) = 0$. Then $(r, 0)$ is an $x$-intercept. The $y$-intercept, if it exists, is found from $f(0)$.

### Reading coordinates

$f(a) = b$ means the point $(a, b)$ lies on the graph. Read an input horizontally and its output vertically.

A graph may also reveal domain, range, symmetry, and how often a horizontal or vertical line intersects it. Endpoints and open/closed points matter.

- zero of a function is the number $r$ that make $f(x)=0$.
- $(r,0)$ is coordinate, is the x-intercept, which is the coordinate that touches x-axis.

# Extract Information from a Formula



Let

$$h(x) = \frac{x^2 - 4}{x - 1}.$$

**a.** Is $(2, 0)$ on the graph of $h$? What about $(-2, 0)$?

**b.** Find the $x$- and $y$-intercepts.

**c.** State the domain in interval notation.

**d.** Evaluate $h(3)$ and interpret the result as a point on the graph.

a). plug in 2, see if 0 is on the graph. $h(2) = \frac{2^2-4}{2-1} = \frac{0}{1} = 0$

same step for $(-2,0)$.

b). x-intercept $\Rightarrow \frac{x^2-4}{x-1} = 0 \Rightarrow x^2-4=0$, solve for x,

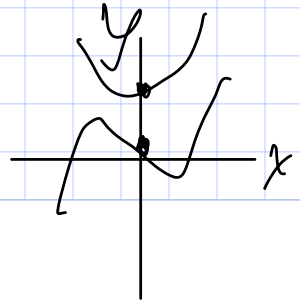
$x^2 = 4$

$x = \pm 2$

$(2,0)$, $(-2,0)$ are x-intercept...

2, -2 are zero of $h(x)$.

y-intercept.




to find y-intercept, plug in 0 for $h(x)$.

$h(0) = \frac{0^2-4}{0-1} = \frac{-4}{-1} = 4$ $(0,4)$ y-intercept.

c. $x-1 \neq 0 \Rightarrow x \neq 1$ $(-\infty, 1) \cup (1, \infty)$

d. ✓

# Read a Piecewise Graph



Use the graph to answer:

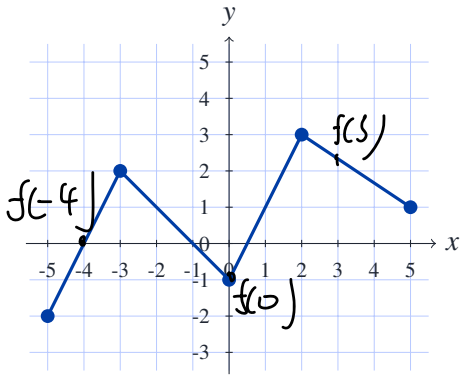
**a.** $f(-4)$, $f(0)$, and $f(3)$

**b.** all $x$- and $y$-intercepts

**c.** the domain and range

**d.** all $x$ such that $f(x) = 2$

**e.** how many times $x = 1$ intersects the graph




b). x-int: $(-4,0)$, $(-2,0)$, $(\frac{1}{2},0)$

y int: $(0,-2)$

c: domain: $[-5,5]$ range: $[-2,3]$

d: $x=-3$ or $1.3$

e: one times.